



E-Commerce Sales Analytics Dashboard

Project Report

Training / Project Institution: SevenMentor, Pune

Power BI project developed as part of Power BI training at SevenMentor, Pune.

1. Project Overview

The E-Commerce Sales Analytics Dashboard is an interactive Power BI project designed to analyze sales, orders, customers, products, payment methods, order status, and regional performance. The dashboard converts raw e-commerce data into meaningful business insights using data cleaning, transformation, data modeling, DAX calculations, interactive filters, and Power BI visualizations.

2. Project Objectives

- Analyze overall sales and order performance.
- Monitor total sales, orders, quantity, and customers.
- Identify high-performing product categories and brands.
- Analyze sales by customer tier.
- Understand delivered, cancelled, and returned orders.
- Analyze payment methods and monthly order trends.
- Analyze customer behavior using gender and customer tier.
- Compare sales performance across states and regions.
- Provide interactive dashboards for business analysis.
- Use Power BI AI capabilities to generate additional business insights.

3. Dashboard Structure

Page 1 – Project Introduction

Acts as the project landing page and provides the project name, navigation, and short descriptions of the analytical dashboard sections.

Page 2 – Executive Sales Overview

Provides high-level sales analysis with filters for Year, Customer Tier, State, Category, and Brand. Displayed KPIs include 606M Total Sales, 25K Total Orders, 32K Total Quantity, and 4K Total Customers.

Page 3 – Order Analytics

Analyzes order processing, delivery, cancellation, return, payment, and monthly order behavior. Displayed KPIs include 250K Total Orders, 4.50 Average Delivery Days, 0.05 Cancellation Rate, 12K Returned Orders, 200K Delivered Orders, 13K Cancelled Orders, and 0.05 Return Rate.

Page 4 – Customer & Product Analysis

Analyzes customer segments, product performance, ratings, gender-based sales, and state-wise sales. Displayed KPIs include 40K Total Customers, 6bn Total Sales, 4.40 Average Rating, 312K Total Quantity, and 23.97K Average Order Value.

4. Data Analysis Process

Data Collection: Imported raw e-commerce datasets into Power BI.

Data Cleaning: Removed unnecessary columns, handled missing values, removed duplicates, corrected data types, standardized categorical values, and formatted dates.

Data Transformation: Used Power Query to transform columns, create required fields, format dates, and prepare analysis-ready data.

Data Modeling: Established relationships between sales/orders, customer, product, date, and location data where applicable.

DAX Calculations: Created measures for Total Sales, Total Orders, Total Quantity, Total Customers, Average Order Value, Average Delivery Days, Cancellation Rate, Return Rate, and Average Rating.

Visualization: Used KPI cards, donut charts, column charts, trend charts, tables, maps, and slicers.

5. Technology Stack

Technology	Purpose
Power BI Desktop	Dashboard development and visualization
Power Query	Data cleaning and transformation
DAX	Measures and business calculations
Excel / CSV	Source data
Data Modeling	Relationships between datasets
Power BI Maps	Geographic sales analysis
Power BI AI Features	Automated insights and narrative analysis

6. Key Business Insights

- Electronics contributes the largest displayed share of category sales on the Executive Sales Overview.
- Platinum customers generate the largest displayed sales contribution among customer tiers.
- Delivered orders form the largest displayed portion of order statuses.
- UPI is the highest-volume payment mode in the displayed Order Analytics view.

- Monthly order volume varies considerably, with June showing a notable peak in the displayed chart.
- The Customer & Product page displays an average rating of 4.40.
- Customer sales can be analyzed by customer tier and gender.
- The geographic visualization enables comparison of sales performance across Indian states.
- Product analysis combines sales, price, brand, and rating information.

7. AI Integration

The project can use Power BI AI capabilities such as Smart Narrative and Key Influencers to support automated interpretation of sales, customer, product, and order patterns.

Example AI-generated insight: Sales are primarily driven by the Electronics category and Platinum customer segment, while order performance varies across payment methods and months.

8. Interactive Features

The dashboard supports interactive filtering and cross-filtering. Users can analyze results by Year, State, Category, Brand, Customer Tier, Gender, Payment Mode, and Order Status.

9. Project Outcome

The project provides a centralized analytical solution for monitoring e-commerce business performance. Users can explore Sales, Orders, Customers, Products, Payments, Returns, Delivery, and Geography through a single interactive Power BI solution.

10. Conclusion

The E-Commerce Sales Analytics Dashboard demonstrates the use of Power BI for business intelligence, including data preparation, transformation, modeling, DAX calculations, interactive visualization, and AI-assisted analysis. The four dashboard pages provide complementary views of executive sales, order operations, and customer/product performance.

11. Training / Project Attribution

This project was developed as part of Power BI training at SevenMentor, Pune. The report includes a SevenMentor-style logo on the cover and identifies the training institution.

Resume Project Summary: Developed an interactive Power BI dashboard to analyze e-commerce sales, orders, customers, products, payments, returns, and regional performance; performed data cleaning, transformation, modeling, DAX calculations, visualization, and AI-assisted analysis during Power BI training at SevenMentor, Pune.